

P-T PROFILE COMPLEXITY IN THE RETRIEVAL OF TRANSMISSIONS SPECTRA

Simon Schleich
Department of Astrophysics, FGGA
University of Vienna

S. Boro Saikia, Q. Changeat, M. Güdel, A. Voigt, I. Waldmann



universität
wien

Department of Astrophysics

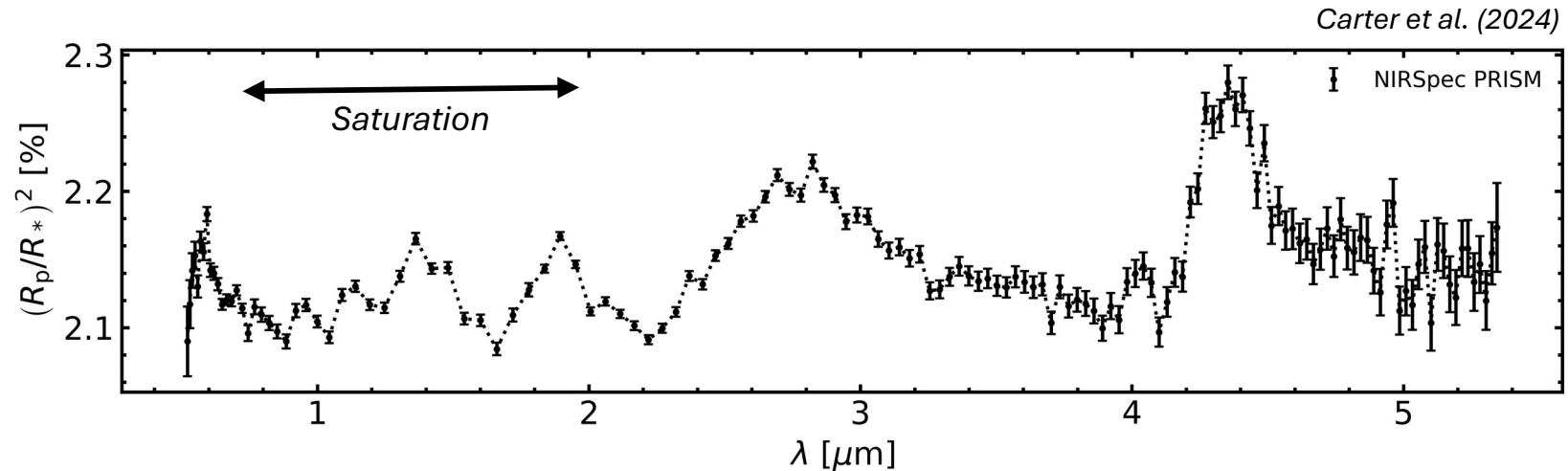
ÖFG //

- Data quality leap with JWST requires appropriately adjusted forward models (e.g. clouds, vertical profiles, ...)
- We know about **increased requirements on p - T profile** for idealised capabilities (e.g. Line+2012,2013; Rocchetto+2016; Blečić+2017)
 - *Indirect influence of p - T profile on transmission spectra*

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- We know about **increased requirements on p - T profile** for idealised capabilities (e.g. Line+2012,2013; Rocchetto+2016; Blečić+2017)
 - *Indirect influence of p - T profile on transmission spectra*
- More **detailed look into individual “observations”**
 - *Complexity threshold? Inherent biases from over-simplified models?*

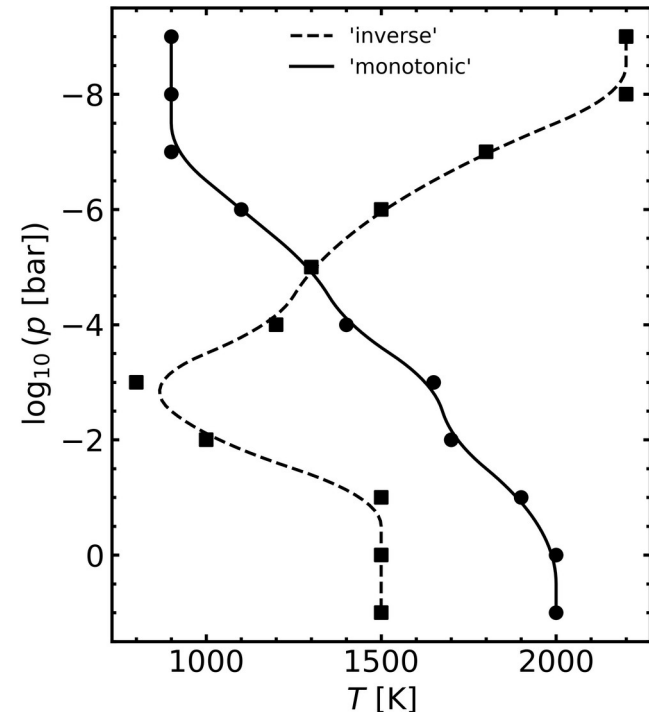
Forward Model Sample

- Synthetic test-spectra modelled after existing observation
 - *WASP-39b, NIRSpec PRISM (excl. saturated region)*
 - *Potential for application onto existing data set*



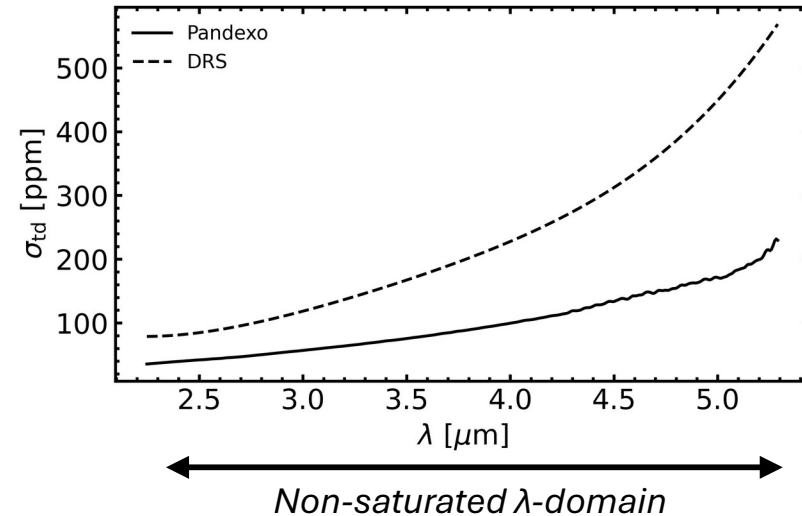
Forward Model Sample

- Synthetic test-spectra modelled after existing observation
- Sample of input cases based on (p, T)
 - *Input profile = 11pt*
 - *Generic profiles of two complexity levels*



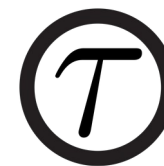
Forward Model Sample

- Synthetic test-spectra modelled after existing observation
- Sample of input cases based on $(p, T) + \sigma_{\text{td}} + p_{\text{cloud}}$
 - “Optimistic” and “conservative” noise case
 - Attenuation by cloud-deck at several pressure-levels



Homogenised retrieval

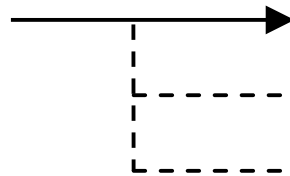
TauREx3 [1]



- Self-retrieval of R_p , $\log_{10}(X_{\text{VMR}})$, $\log_{10}(p_{\text{cloud}})$
- Complexity variation for heuristic p - T profiles

Input

p - T profile (11 parameters)
 (p_n, T_n) –pairs



Retrieval

“Free” p - T profiles

- *Isothermal* $\rightarrow$ 8-point
- 11-point (validation)

Evaluation

Bayes' factor



Model preference based
on trade-off between
fit-quality and
parameter number

Parameter accuracy



A good fit might be
produced by settling
on the wrong parameters

p-T accuracy



With a known
ground-truth, retrieved
p-T profiles can be
compared



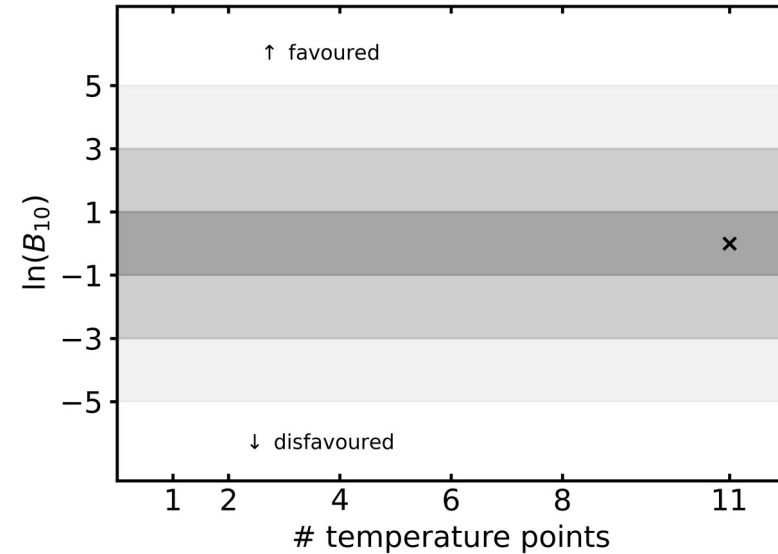
6 cases of p-T complexity

Bayes' factor

Parameter accuracy

p-T accuracy

- Evaluation relative to validation run

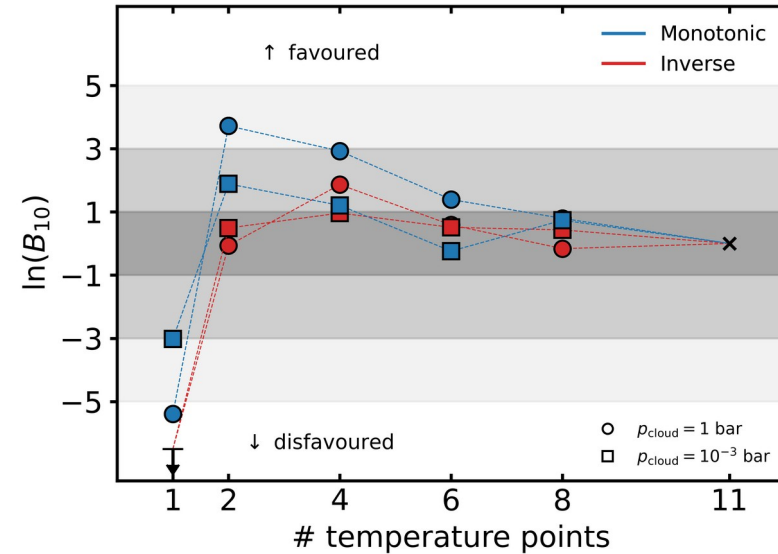


Bayes' factor

Parameter accuracy

p-T accuracy

- Evaluation relative to validation run

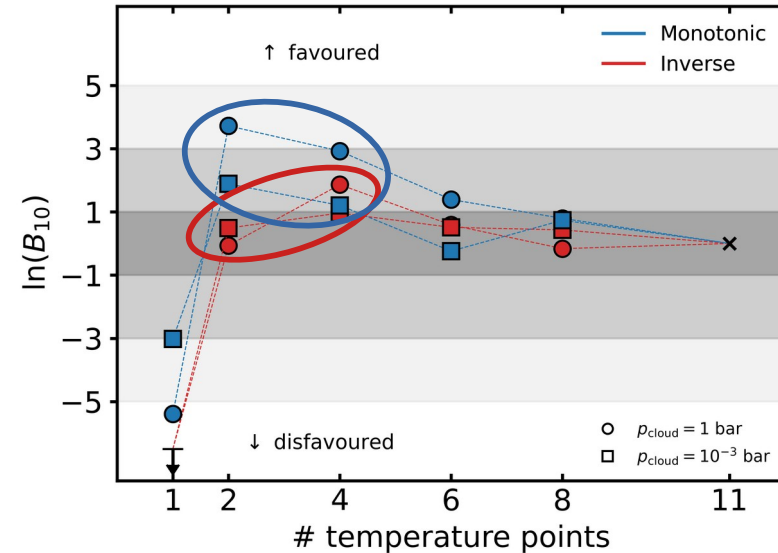


Bayes' factor

- Evaluation relative to validation run
- Preference dependent on underlying p-T structure (around 2- to 4-point)

Parameter accuracy

p-T accuracy

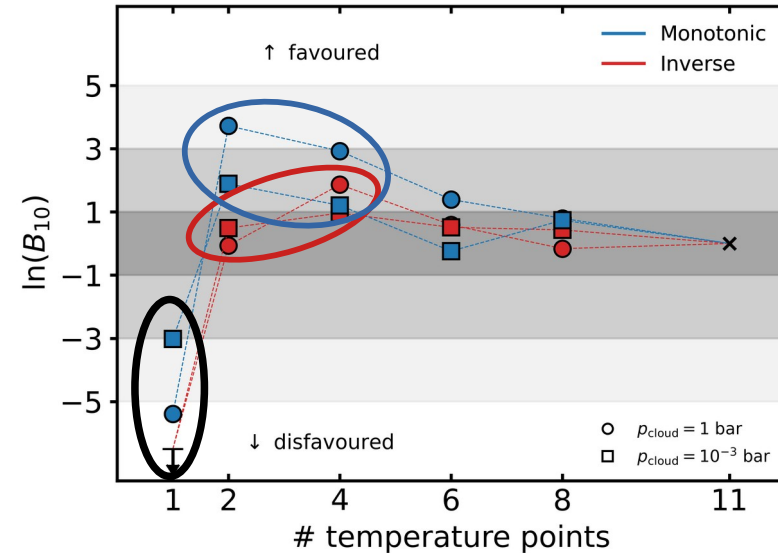


Bayes' factor

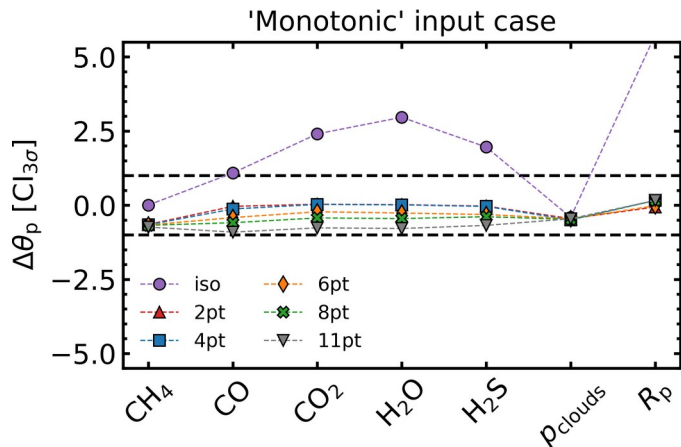
- Evaluation relative to validation run
- Preference dependent on underlying p-T structure (around 2- to 4-point)
- Clearly disfavoured isothermal case

Parameter accuracy

p-T accuracy



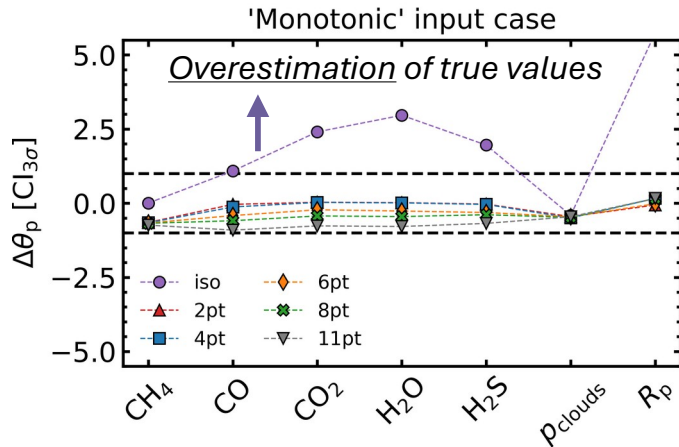
Evaluation – II



Parameter accuracy

- Strong bias in isothermal retrieval
- Equivalent performance of other profiles

p-T accuracy

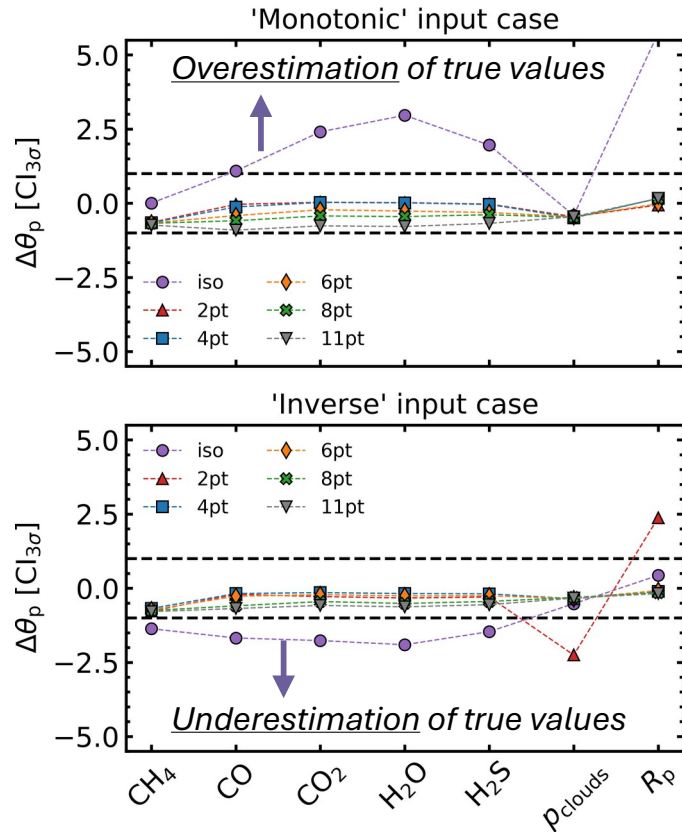


Parameter accuracy

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p-T accuracy

Evaluation – II

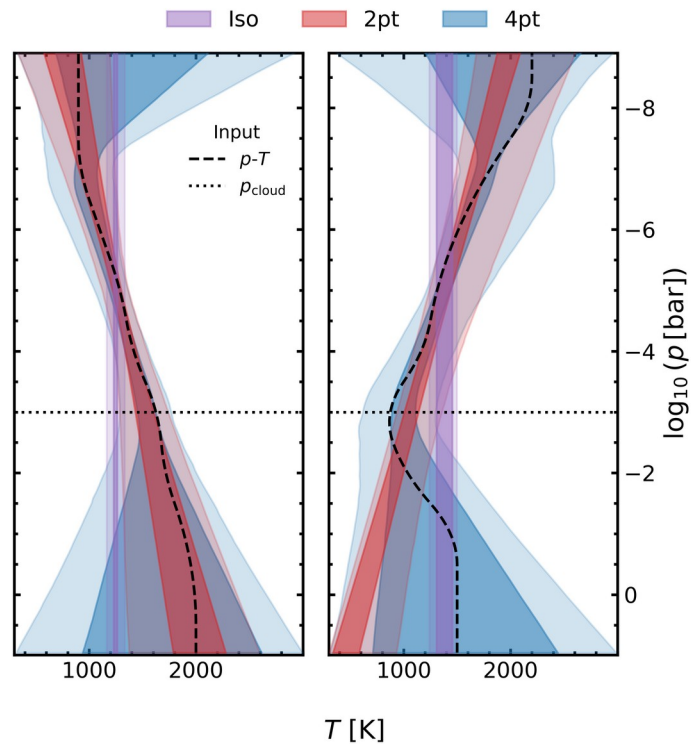
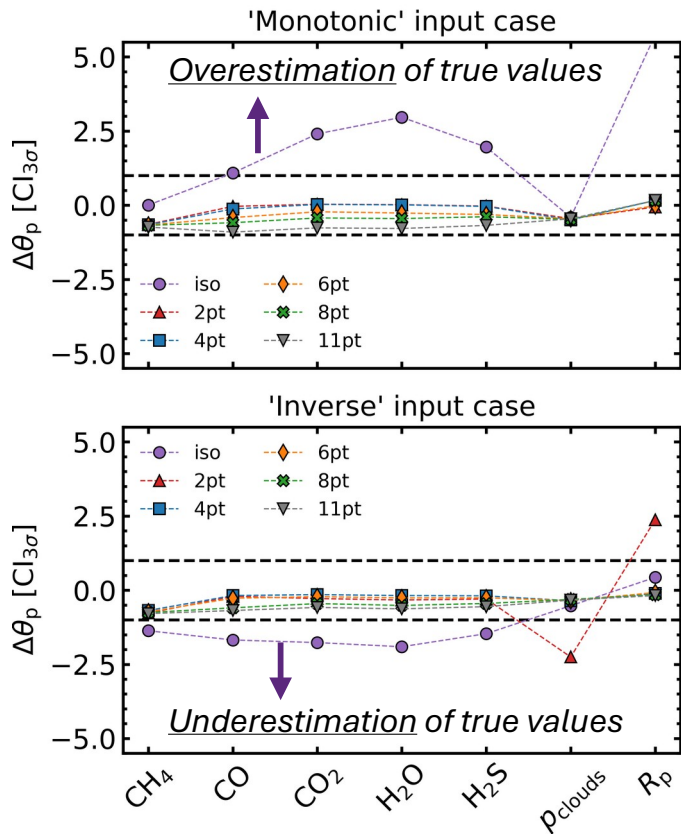


Parameter accuracy

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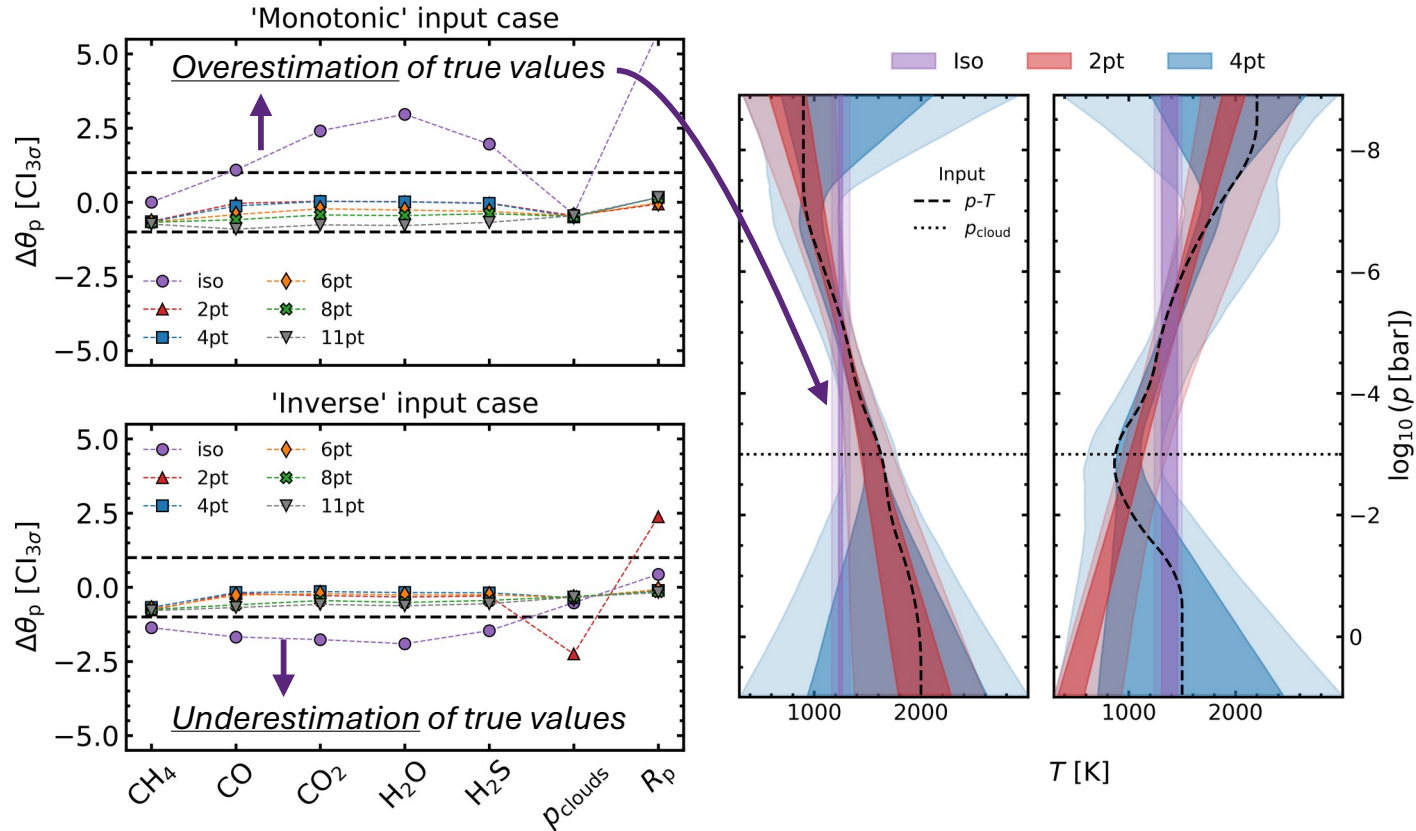
p-T accuracy

Evaluation – III



p-T accuracy

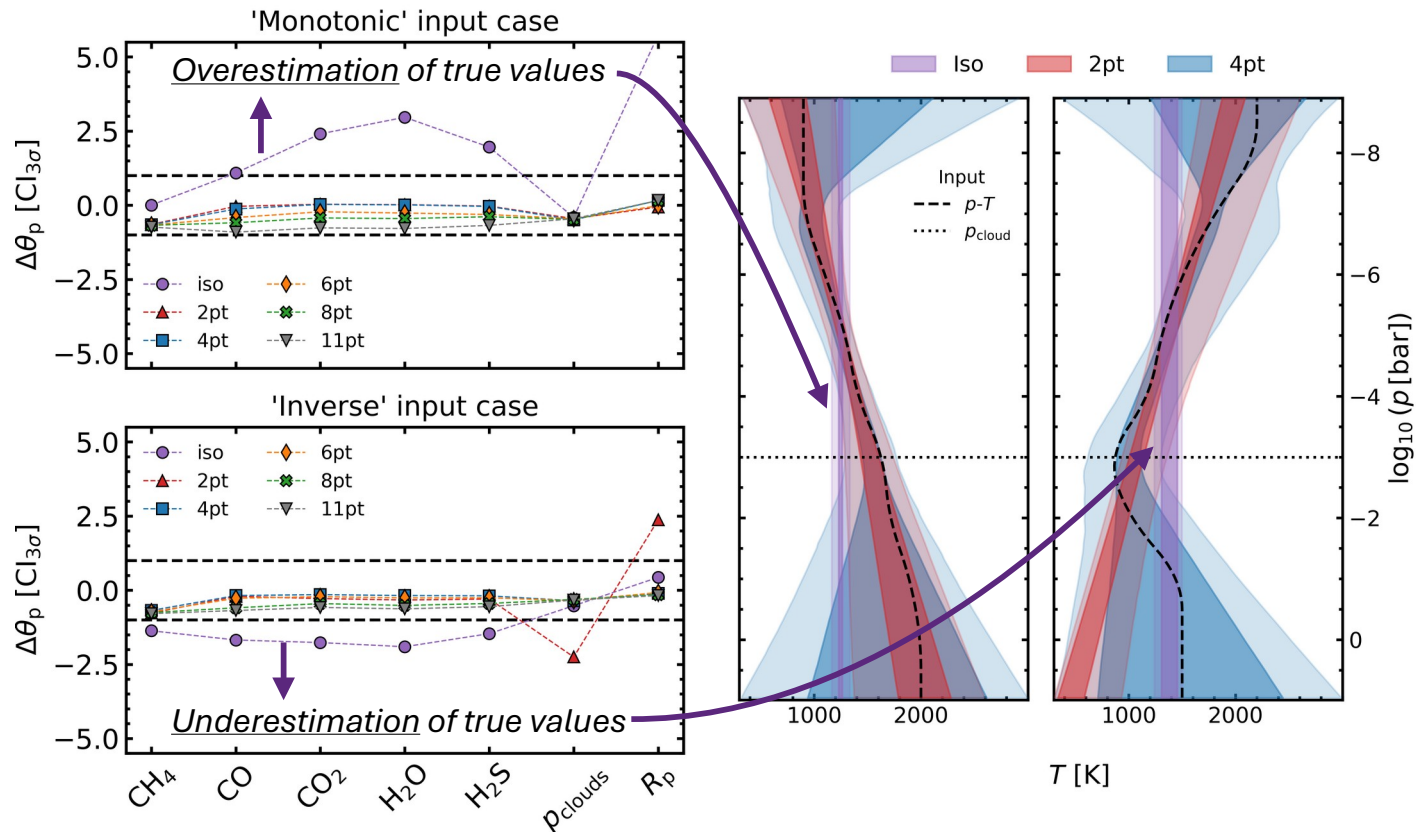
Evaluation – III



p-T accuracy

- Anti-correlated with retrieved temperature
- Scale height $\rightarrow$ path length in optical depth

Evaluation – III



p-T accuracy

- Anti-correlated with retrieved temperature
- Scale height → path length in optical depth

Conclusion

- Non-isothermal, variable preference based on underlying p - T structure
- JWST data quality warrants systematic investigations into model complexity
- Evaluation of actual observations combined with forward model investigations



Additional Slides

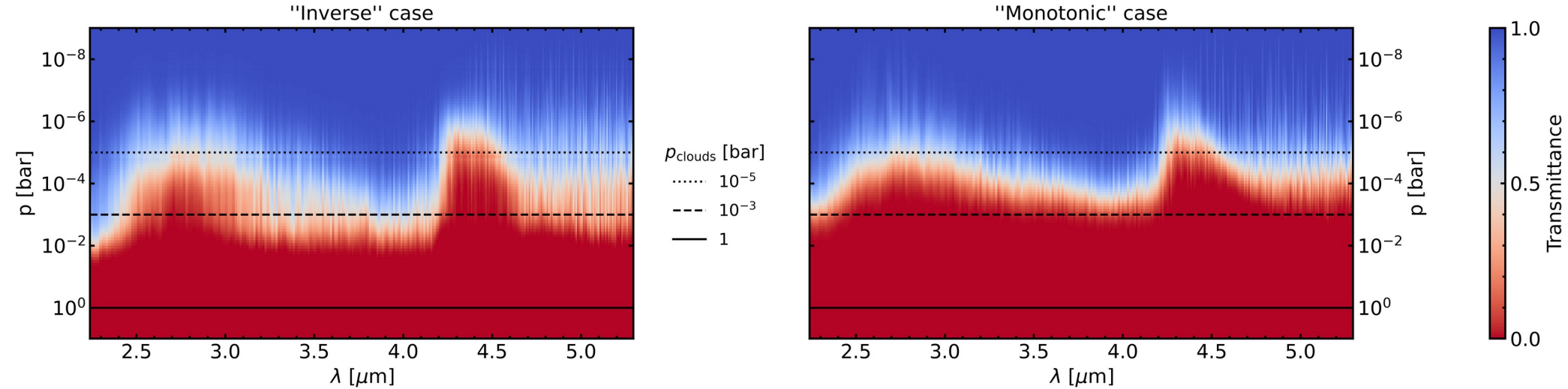
System Parameters

Parameter	Value	Unit
Star		
M_*	0.918 ± 0.047	$M_\odot$
R_*	1.013 ± 0.022	$R_\odot$
T_{eff}	5485 ± 50	K
[Fe/H]	0.01 ± 0.09	dex
$\log g$	4.41 ± 0.15	dex
Planet		
M_p	0.281 ± 0.032	M_j
R_p	1.279 ± 0.040	R_j
P	$4.0552941 \pm 3.4 \times 10^{-6}$	d

Homogeneous Chemistry

Parameter	Value
He/H ₂	0.13
$X_{\text{H}_2\text{O}}$	5×10^{-3}
X_{CO}	4×10^{-3}
$X_{\text{H}_2\text{S}}$	5×10^{-4}
X_{CO_2}	5×10^{-5}
X_{CH_4}	1×10^{-7}

Optical depth and transmittance



$$\tau_i(\lambda, z) = \int_z^{z_{\text{max}}} \sigma_{i,\lambda}(\hat{z}) \chi_i(\hat{z}) \rho(\hat{z}) d\hat{z}$$

$$\mathcal{T} = \exp(-\tau) \quad \rightarrow \quad \text{Transmittance}$$

- Blečić, J., Dobbs-Dixon, I., & Greene, T. 2017, ApJ, 848, 127
- Line, M. R., Zhang, X., Vasisht, G., et al. 2012, ApJ, 749, 93
- Line, M. R., Wolf, A. S., Zhang, X., et al. 2013, ApJ, 775, 137
- Rocchetto, M., Waldmann, I. P., Venot, O., et al. 2016, ApJ, 833, 120

[1] TauREx3.1 based on:

- Al-Refaie, A. F., Changeat, Q., Venot, O., et al. 2022, ApJ, 932, 123
- Al-Refaie, A. F., Changeat, Q., Waldmann, I. P., & Tinetti, G. 2021, ApJ, 917, 37
- Waldmann, I. P., Tinetti, G., Rocchetto, M., et al. 2015, ApJ, 802, 107